WORKED SOLUTION

STEP I 2006, Question 9

Mechanics • Connected Particles

STEP I 2006, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q9

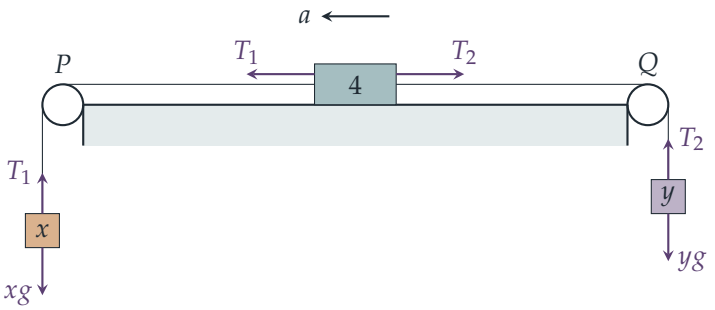
The key idea

The motion splits into two stages, each with constant acceleration. While both strings are attached, the whole system of three bodies moves as one, driven by the difference of the two hanging weights. Once the string is cut, only the lighter particle pulls on the block, now against the motion, so the block decelerates uniformly. The speed at the moment of cutting is the link between the two stages, and the constraint $x + y = 6$ lets us write everything in terms of y.

Solution

Model

We treat the block and the two hanging masses as particles, the strings as light and inextensible, and the table and pulleys as smooth. Since $x > y$, the block moves towards P, the particle of mass x descends and the particle of mass y rises. Let T_1 and T_2 be the tensions in the strings over P and Q.



Stage 1: both strings attached

Let the common acceleration be a_1 (block towards P, particle x down, particle y up). Newton's second law for each body gives

$$\begin{aligned} xg - T_1 &= xa_1, \\ T_1 - T_2 &= 4a_1, \\ T_2 - yg &= ya_1. \end{aligned}$$

Adding eliminates the tensions:

$$(x-y)g = (x+y+4)a_1 = 10a_1, \qquad \text{so} \qquad a_1 = \frac{(x-y)g}{10} = \frac{(6-2y)g}{10},$$

using $x = 6 - y$. This is positive because $y < x$ means $y < 3$.

Starting from rest with constant acceleration, the block covers distance d in time t_1 and reaches speed V, where

$$d = \tfrac{1}{2}a_1 t_1^2, \qquad V^2 = 2a_1 d.$$

Hence

$$t_1 = \sqrt{\frac{2d}{a_1}} = \sqrt{\frac{20d}{(6-2y)g}} = \sqrt{\frac{d}{5g}}\,\frac{10}{\sqrt{6-2y}}, \qquad V = \sqrt{\frac{(6-2y)gd}{5}}.$$

Stage 2: after the cut

Now $T_1 = 0$. The block is still moving towards P with speed V, and the particle of mass y is still rising with speed V. Only the string over Q acts on the block, and it pulls back towards Q. Let the deceleration be a_2. Then

$$\begin{aligned} T_2 &= 4a_2 \quad \text{(block)}, \\ yg - T_2 &= ya_2 \quad \text{(particle } y\text{, decelerating while rising)}, \end{aligned}$$

so

$$a_2 = \frac{yg}{4+y}.$$

Ask yourself How do we know the string over Q stays taut after the cut?

If the string went slack, the particle would decelerate at g while the block would keep its speed. With the string taut the tension is $T_2 = 4a_2 = 4yg/(4+y) > 0$, which is consistent, and the common deceleration a_2 is less than g. So the string really does stay taut, and the block and particle move together until they stop.

The block comes to rest after a further time

$$\begin{aligned} t_2 = \frac{V}{a_2} &= \sqrt{\frac{(6-2y)gd}{5}} \cdot \frac{4+y}{yg} \\ &= \sqrt{\frac{d}{5g}}\,\sqrt{6-2y}\,\left(1 + \frac{4}{y}\right), \end{aligned}$$

where we wrote $\sqrt{(6-2y)gd/5}\,/\,g = \sqrt{d/(5g)}\sqrt{6-2y}$ and $(4+y)/y = 1 + 4/y$.

Total time

The block first returns to rest at the end of stage 2, so the total time is

$$t_1 + t_2 = \sqrt{\frac{d}{5g}}\left[\frac{10}{\sqrt{6-2y}} + \left(1+\frac{4}{y}\right)\sqrt{6-2y}\right] = \sqrt{\frac{d}{5g}}\,\mathrm{f}(y),$$

as required.

The stationary value of f

The physical range is $0 < y < 3$. Write $u = 6 - 2y$, so $\mathrm{d}u/\mathrm{d}y = -2$ and

$$\mathrm{f}(y) = 10u^{-1/2} + \left(1+\frac{4}{y}\right)u^{1/2}.$$

Differentiating, with the product rule on the second term,

$$\begin{aligned}\mathrm{f}'(y) &= 10\cdot\left(-\tfrac{1}{2}\right)u^{-3/2}(-2) - \frac{4}{y^2}u^{1/2} + \left(1+\frac{4}{y}\right)\tfrac{1}{2}u^{-1/2}(-2)\\ &= \frac{10}{u^{3/2}} - \frac{4u^{1/2}}{y^2} - \frac{y+4}{y\,u^{1/2}}.\end{aligned}$$

Multiplying by $y^2u^{3/2}$, which is positive for $0 < y < 3$, the condition $\mathrm{f}'(y) = 0$ becomes

$$10y^2 - 4u^2 - (y^2+4y)u = 0.$$

Substituting $u = 6 - 2y$:

$$\begin{aligned}10y^2 - 4(36 - 24y + 4y^2) - (6y^2 - 2y^3 + 24y - 8y^2) &= 0\\ 2y^3 - 4y^2 + 72y - 144 &= 0\\ 2(y-2)(y^2+36) &= 0.\end{aligned}$$

Since $y^2 + 36 > 0$, the only solution is

$$y = 2,$$

which lies in $0 < y < 3$ (and gives $x = 4$).

Because $\mathrm{f}(y) \to \infty$ as $y \to 0^+$ and as $y \to 3^-$, this single stationary point is the minimum: $\mathrm{f}(2) = 10/\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$, so the shortest possible time to rest is $8\sqrt{2d/(5g)}$.

Checks

- Stepping the motion numerically (time step 2×10^{-6} s) for random $y \in (0,3)$ and d reproduced $\sqrt{d/(5g)}\,\mathrm{f}(y)$ to five significant figures; for example $y = 1.671$, $d = 0.581$ gives 1.27096 s both ways with $g = 9.8$.
- A numerical derivative of f changes sign exactly once on $(0,3)$, at $y = 2$, and $\mathrm{f}(2) = 11.3137\ldots = 8\sqrt{2}$.

- Limiting cases make sense: as $y \to 3$ the masses nearly balance, $a_1 \to 0$ and stage 1 takes forever; as $y \to 0$ nothing slows the block down after the cut, so stage 2 takes forever.

Common slips

- Using mass $4 + x + y$ in stage 2: once the string is cut, the particle of mass x is no longer part of the system.
- Forgetting that the particle of mass y is still moving upwards at speed V after the cut, and so treating stage 2 as starting from rest.
- Dropping $\mathrm{d}u/\mathrm{d}y = -2$ when differentiating $\sqrt{6 - 2y}$.

Beyond STEP

Splitting a motion into stages joined by continuity of velocity is the basic idea behind piecewise-defined dynamics, which reappears in impulsive forces and in control problems where a force is switched on or off. Asking which choice of y minimises the time is a first taste of optimisation in mechanics, which grows into the calculus of variations (the brachistochrone is the classic example). Kleppner and Kolenkow, *An Introduction to Mechanics*, treats connected bodies and constraint forces carefully.

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